

Initial Literature Review Plan

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1. Topic

This review examines how Vision-Language Models (VLMs) and multimodal architectures have been applied to spatiotemporal reasoning for safety monitoring and activity tracking on active construction sites.

2. Research Questions and Purpose

Purpose: To systematically map the current state of VLM research applied to spatiotemporal reasoning in construction—identifying prevalent architectures, data fusion strategies, deployment barriers, and evaluation frameworks—and to locate the gaps where future primary research is most needed.

Research Questions:

RQ1: What are the prevalent VLM architectures currently utilized for spatiotemporal analysis in construction environments?

RQ2: How is multimodal data—visual and textual—fused and processed to evaluate safety hazards on active construction sites?

RQ3: What are the primary methodological bottlenecks in deploying these models for real-time progress tracking and safety monitoring?

RQ4: How is the performance of VLMs currently evaluated and benchmarked for spatiotemporal tasks in construction contexts?

3. Selected Review Methodology

Methodology: Scoping review, structured according to PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines. This methodology maps the breadth and nature of available evidence— architectures, datasets, and evaluation practices—without requiring methodological homogeneity across included studies. It is appropriate for an emerging, multidisciplinary field in which evaluation metrics and study designs vary substantially across the literature, making statistical meta-analysis premature.

4. Search Strategy

Keyword development. Search keywords were identified by analyzing terminology used in recent studies that apply artificial intelligence to construction management and safety. The search query was then structured using a tripartite Boolean logic framework designed to intersect three distinct conceptual domains: (1) the specific AI architecture (multimodal vision-language systems); (2) the analytical task (spatiotemporal and dynamic reasoning); and (3) the application domain (construction management and safety).

Search strings applied across both databases:

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String 1 - AI Architecture:
("Vision-Language Model*" OR "VLM" OR "Vision-Text"
OR "Image-Text" OR "Visual Question Answering" OR "VQA"
OR "Image Captioning" OR "CLIP" OR "Large Multimodal Model*"
OR "LMM" OR "Foundation Model*" OR "Generative AI"
OR "Video-Language" OR "Video-Text")

AND

String 2 - Analytical Task:
("Spatiotemporal" OR "Spatio-temporal" OR "Video"
OR "Temporal" OR "Dynamic" OR "Time-series" OR "4D"
OR "Motion" OR "Tracking" OR "Sequenc*")
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OR "Action recognition" OR "Activity recognition")

AND

String 3 - Application Domain:
("Construction Management" OR "Civil Engineering" OR "AEC"
OR "Infrastructure" OR "Built Environment"
OR "Construction Site" OR "Safety Monitoring"
OR "Structural Health Monitoring" OR "Digital Twin")

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Refinement plan. After running the initial string in each database and recording hit counts, result relevance will be evaluated. If the hit set is too broad—for example, dominated by general computer vision papers with no construction context—subject-area filters (engineering, civil engineering, computer science) will be applied within the database interface. If the hit count is too low, additional synonyms will be introduced within the relevant cluster (e.g., **jobsite**, **worksite** alongside **Construction Site**). The search is also supplemented by a **backward snowball phase**—examining reference lists of included full-text papers—to capture foundational computer vision and early multimodal studies that predate the core VLM search window but remain essential comparative baselines.

Anticipated challenges. Because this topic spans construction management, computer vision, and natural language processing, relevant papers may be distributed across journals that do not fully overlap in any single database. Terminology is also inconsistent across disciplines: the same class of model may appear as a **vision–language model**, **multimodal transformer**, or **large multimodal model** in different papers. The broad synonym coverage in String 1 is designed specifically to address this risk. All synonym expansions and any deviations from the original strings will be documented throughout the process.

5. Databases and Sources

The primary databases are **Web of Science** (Core Collection) and **Scopus**. These two platforms were selected because together they provide broad, multidisciplinary coverage of

high-impact peer-reviewed journals spanning civil engineering, construction management, computer science, and applied AI—the four disciplines most directly relevant to this review.

Web of Science is particularly strong in journals indexed in the Science Citation Index Expanded (SCIE), which includes leading safety science, automation, and engineering venues. Scopus complements this with wider coverage of applied engineering and construction management journals, and is frequently cited in scoping review methodology as a preferred companion database to Web of Science. Running the identical search string across both platforms in parallel reduces the risk that relevant literature indexed in only one database will be missed—a recognized best practice in evidence synthesis.

Google Scholar will not serve as a primary database because its results are not consistently reproducible across sessions, which conflicts with the PRISMA-ScR transparency requirements. Preprint servers such as arXiv are also excluded, as the review is restricted to high-impact peer-reviewed publications.

6. Inclusion and Exclusion Criteria

The eligibility criteria below are organized along four dimensions that directly correspond to the scope implied by the four research questions. These criteria are applied sequentially at both the title-and-abstract screening stage and the full-text review stage.

Dimension	Inclusion Criteria	Exclusion Criteria
Publication Date	2019–2026 for core VLM literature; 2010–2026 for foundational computer vision baselines.	Pre-2010; post-March 2026.
Source Type	High-impact, peer-reviewed journal articles.	Editorials, book chapters, theses, non-English documents, conference proceedings, and industry white papers.
Technology	Multimodal architecture integrating visual/geometric data with textual/semantic data (e.g., VLMs, VQA, Image Captioning, CLIP-based models, Large Multimodal Models).	Purely unimodal computer vision (e.g., standalone YOLO, CNN bounding-box detectors) or purely text-based NLP, unless serving as an explicit comparative baseline within a hybrid multimodal framework.
Application Domain	Active construction phase: Safety Management (hazard identification, PPE compliance) and Progress Tracking (productivity analysis, schedule monitoring, reporting).	Post-construction or facility management contexts, structural health monitoring outside the active construction phase, and non-engineering domains.

Screening procedure. Screening will follow a two-stage PRISMA-ScR process. Stage 1 screens titles and abstracts to remove clearly out-of-scope records. Stage 2 applies the full eligibility criteria to remaining full texts, with one standardized exclusion reason recorded per rejected paper (e.g., **wrong domain, non-peer-reviewed, no temporal component, purely unimodal**). Record counts at each stage will be reported in a PRISMA-ScR flow diagram included in the final submission.

7. Reflection on the Search Process

Three developments have meaningfully shaped this plan since the initial proposal. First, in response to instructor feedback from Week 1, the original broad framing—VLMs in built environments generally—was narrowed to focus specifically on safety monitoring and activity tracking during the active construction phase; this change made the search terms, database choices, and eligibility criteria substantially more coherent and focused. Second, a single broad research question was restructured into four operationally distinct questions (RQ1–

RQ4) that define clear, non-overlapping mapping targets: the Technology inclusion criterion now directly operationalizes the scope implied by RQ1 and RQ2, while the Application Domain criterion reflects the construction-specific focus required by all four questions. Third, the adoption of PRISMA-ScR as the reporting standard formalized what were initially informal methodological decisions, and the inclusion of a backward snowball phase emerged from recognizing that foundational computer vision baselines (2010 onward) often predate the keyword-indexed VLM literature and would otherwise be missed by the primary database search alone.